Hypoglycemia-induced VEGF expression is mediated by intracellular Ca^{2+} and protein kinase C signaling pathway in HepG2 human hepatoblastoma cells

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Abstract. Vascular endothelial growth factor (VEGF) is a potent angiogenic factor that plays a central role in angiogenesis. In this study, we investigated the mechanism of VEGF expression in HepG2 human hepatoblastoma cells under hypoglycemia. The shortage of glucose significantly enhanced VEGF mRNA expression in a time-dependent manner as well as increased DNA-binding activity of AP-1 that plays an important role in VEGF transcription. In addition, treatment of a potent PKC inhibitor, H-7 in glucose-deprived HepG2 cells suppressed hypoglycemia-elevated VEGF expression as well as the increased AP-1 DNA-binding activity. Moreover, we observed that Ca^{2+} levels remarkably increased under low glucose condition. Consistently, an intracellular Ca^{2+} chelator, BAPTA/AM significantly decreased hypoglycemia-induced VEGF expression and AP-1 DNA-binding activity. Therefore, these results indicate that increase of intracellular Ca^{2+} level induces the activation of PKC, which induce the activation of AP-1 leading to the increase of VEGF in glucose-deprived environment. Furthermore, it provides one link in regulation of VEGF with hypoglycemia as well as information to understand how hypoglycemia induces VEGF expression and subsequently leads to tumor angiogenesis.

Introduction

Physiologically, angiogenesis, formation of new capillary blood vessels, is essential for embryogenesis, development, ovulation, corpus luteum formation, and wound repair (1,2). Angiogenesis is an early event in tumor development and metastasis and mediated by several angiogenic factors such as basic fibroblast growth factor (bFGF), angiogenin, platelet derived growth factor (PDGF), transforming growth factor α and ß (TGF α and ß), and vascular endothelial growth factor (VEGF) (3). These factors can be induced by environmental stress like hypoxia or hypoglycemia (3), which leads to cancer cells with more nutrients through the formation of new blood vessels.

VEGF is a secreted, 46-kDa dimeric protein with structural homology to placental derived growth factor (PlGF) (4). As an endothelial cell-specific mitogen, VEGF plays key roles in normal and abnormal neovascularization. Transcriptional factors like AP-1, AP-2 or SP1 tightly control VEGF expression which has numerous and putative binding sites in its promoter region (5). Its expression is up-regulated by hypoxia (6), hypoglycemia (7), epidermal growth factor (EGF) (8), estrogen (9,10), and cobalt chloride (11), however, it remains largely unknown how hypoglycemia regulates VEGF expression.

Stress-induced AP-1 DNA-binding activity is due to the post-translational modification of both newly synthesized and pre-existing Jun and Fos proteins (12). Several experiments demonstrate that transcriptional activation of the c-*jun* gene is exerted through the distal and proximal AP-1 binding sites in its promoter (13). Studies by Datta *et al* (14) suggest that stress-induced transcriptional activation of the c-*jun* gene is mediated through the formation of reactive oxygen intermediates, and that PKC is also involved in the signal pathway. PKC is a family of protein-serine kinases and has been implicated in regulation of many cellular functions. PKC is transcriptionally activated by c-*jun* (14) resulting in not only increasing intraocular vascular permeability *in vivo* (15) but also up-regulating VEGF expression in medial smooth muscle cells of the arterial wall (16).

The concentration of Ca^{2+} is considered as an important regulator of diverse cell functions (17) and it modulates activities of many enzymes including cyclic AMP phosphodiesterase (18), phospholipase C (19), and PKC (20). Intracellular Ca^{2+} level is markedly increased in hypoglycemic condition in brain slices (21) and directly involved in signal

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Abbreviations: H-7, 1-(5-isoquinolinylsulfonyl)-2-methylpiperazine; PKC, protein kinase C; VEGF, vascular endothelial growth factor

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transduction for VEGF induction (16). Therefore, it is speculated that induction of VEGF expression by hypoglycemia may be closely related to intracellular Ca^{2+} levels, PKC and transcription factor AP-1.

We demonstrate that hypoglycemia induces both VEGF expression and AP-1 DNA-binding activity in HepG2 cells and that an increase of intracellular Ca^{2+} levels and activation of PKC are significantly involved in hypoglycemic induction of VEGF.

Materials and methods

Cell culture. HepG2 cells (1x10^6 cells) were plated onto 100-mm culture dishes and grown in RPMI-1640 medium (Life Technology, Gaithersberg, MD) supplemented with 10% FBS (Life Technology), penicillin (100 µU/ml), and streptomycin (100 µg/ml). For condition of hypoglycemia, cells were rinsed with PBS three times, and cultured in glucose-free RPMI-1640 medium containing 10% FBS. Treatment of BAPTA/AM (Molecular Probes, OR) or H-7 (Sigma Chemical Co., St. Louis, MO) was accomplished by aspirating the media and replacing it with fresh media.

Measurement of intracellular Ca^{2+} levels. Aliquots of HepG2 cells were cultured for 3-5 days and washed in Eagle's basal salt solution (EBSS). Two µM of Fura-2/AM were added and cells were incubated for 60 min at room temperature (22-23˚C). Unloaded Fura-2/AM was discarded by centrifugation for 3 min at 150 x g. Cells were resuspended in Ca^{2+}-free Krebs-Ringer Buffer (KRB) (125 mM NaCl, 5 mM KCl, 1.2 mM KH_2PO_4, 1.2 mM $MgSO_4$, 5 mM $NaHCO_3$, 25 mM HEPES, pH 7.4) with or without 6 mM glucose, and transferred to a quartz cuvette and continuously stirred. Fluorescence emission (510 nm) was monitored with excitations in wavelength between 340 and 380 nm at 37˚C using a Hitachi F2000 fluorimeter. Maximum and minimum values of fluorescence at each excitation wavelength were obtained after lysing the cells with 0.1% Triton X-100 (maximum), or after adding 10 mM EGTA (minimum). With the maximum and minimum values, the fluorescence ratios of 340:380 nm were converted into free Ca^{2+} concentration using the Fura-2 Ca^{2+}-binding constant (224 nM) and the formula described by Grynkiewicz *et al* (22).

Northern blot analysis. Total RNA was isolated using Trizol (Life Technology, Gaithersberg, MD) according to the manufacturer's instructions. For Northern blot analysis, 30 µg of total RNA was electrophoresed in 1% agarose-formaldehyde gels. The RNA was blotted onto nylon membranes (BioRad Lab., Hercules, CF), and hybridized with [α-^{32}P]dCTP-labeled 0.5-kb human VEGF cDNA fragment. The autoradiographic films were scanned with a computerized laser scanning densitometer (Molecular Dynamics, Suunyvale, CA) for comparing relative band density.

Preparation of nuclear proteins. Cells were washed with cold PBS three times and scraped. After collection of cells by centrifugation at 2,000 rpm for 5 min, the pellet was resuspended in buffer A (10 mM HEPES, 1.5 mM $MgCl_2$, 0.25% Nonidet P-40, pH 7.5) and incubated on ice for 5 min, followed by cetrifugation at 4,000 rpm for 2 min. The supernatant (the cytosol) was removed and the nuclei were extracted with buffer C (20 mM HEPES, 25% glycerol, 420 mM NaCl, 0.2 mM EDTA, 0.25% Nonidet P-40, pH 7.5). The nuclear extracts were vortexed vigorously several times over 20 min, followed by centrifugation at 14,000 rpm for 5 min. The supernatant (nuclear extract) was diluted 1:2 with buffer D (20 mM HEPES, 50 mM KCl, 0.2 mM EDTA, 20% glycerol) and kept frozen at -80˚C until use.

Gel mobility shift assay. The double strand oligonucleotides containing the TRE sequences (Promega, Co., Maison, WI) was 5'-end labeled with [γ-^{32}P]-ATP using T4 polynucleotide kinase and purified by ethanol precipitation. Binding reactions were carried out in a buffer containing 10 mM Tris-HCl (pH 7.5), 50 mM NaCl, 0.5 mM DTT, 10 mM $MgCl_2$, 10% glycerol, 0.05% NP-40, 2 µg of poly (dI-dC) on ice for 15 min. The assay mixture was incubated with radiolabeled oligonuleotides for 30 min at room temperature. After adding 6X dye solution (0.1% bromophenol blue, 30% glycerol), the mixture was immediately loaded and electrophoresed on a non-denaturing 6% polyacrylamide gel in 0.25X TBE (1X TBE: 89 mM Tris, 89 mM boric acid and 2.5 mM EDTA) for 2 h at 200 V. The gels were then dried in a vacuum dryer at 80˚C for 1.5 h and autoradiographed on Fuji RX X-ray film. For supershift experiment, nuclear extracts were incubated with 2 µg of anti-c-Jun antibody for 20 min prior to the addition of ^{32}P-labeled oligonucleotides.

Results

Hypoglycemia increases VEGF expression and AP-1 DNA-binding activity. To determine whether hypoglycemia can induce VEGF expression, we isolated total RNA from HepG2 cells in glucose-free medium with various incubation times and carried out Northern blot analysis. As shown in Fig. 1A, HepG2 cells exposed to hypoglycemia remarkably increased VEGF mRNA in a time-dependent manner. The maximum expression was shown about 6-fold increase at 16 h and diminished significantly until 48 h (data not shown). However, the level of VEGF sustained higher than that under normal condition. Since it has been shown that c-Jun and c-Fos are increased by hypoglycemia in MCF7/ADR multidrug-resistant human breast carcinoma cells (23), we tried to reveal AP-1 activity in hypoglycemic condition. After incubating HepG2 cells for 1 h with glucose-free condition, we performed EMSA with isolated nuclear proteins. As a result, hypoglycemia enhanced AP-1 DNA-binding activity in HepG2 cells, indicating that VEGF induction by the absence of glucose might be mediated through AP-1 (Fig. 1B). To show the specific binding of AP-1, we applied anti-c-Jun antibody on nuclear extracts prior to adding labeled AP-1 oligonucleotides (Fig. 1B, lane 5, supershifted band). These observations suggest that the increased AP-1 activity by hypoglycemia may transactivate the VEGF gene.

Protein kinase C (PKC) plays a role on hypoglycemia-induced VEGF expression and AP-1 DNA-binding activity. PKC mediates VEGF induction (15), and hypoglycemia increases PKC activity during expression of bFGF. In addition,

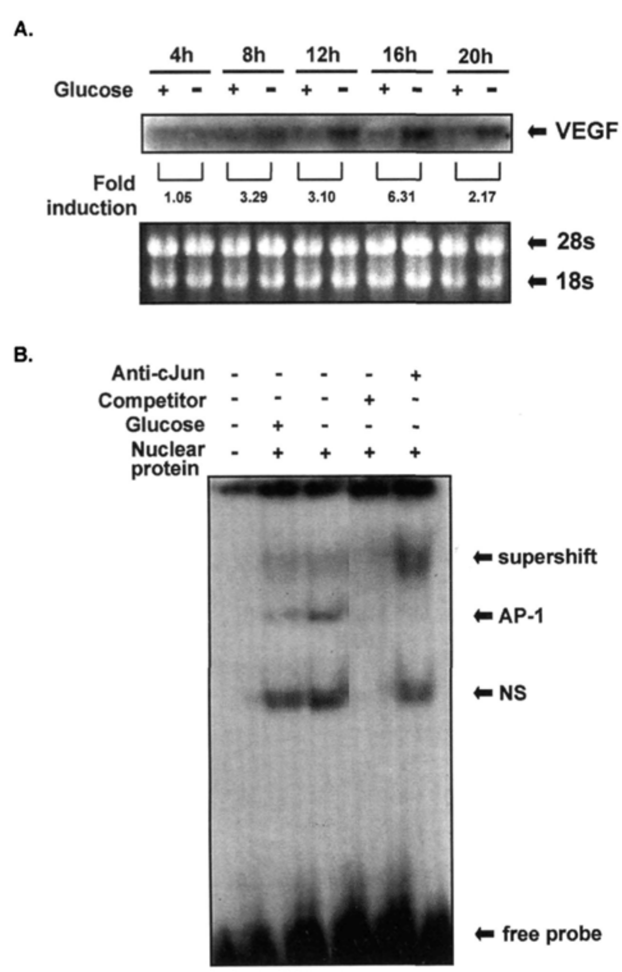

Figure 1. Hypoglycemia induced VEGF mRNA expression and AP-1 DNA-binding activity. A, HepG2 cells were incubated in normal or glucose-free medium for the indicated times (4-20 h). Total RNA (30 µg) from cells was electrophoresed and Northern blot analysis was performed. The autoradiogram was analyzed with a densitometer. Equal amount of RNA was visualized by ethidium bromide staining. B, after incubating cells for 1 h under normal or glucose-deprived condition, nuclear proteins were isolated and EMSA was performed. The competition assay was carried out with addition of 200-fold molar excess of non-labeled AP-1 oligonucleotides. Supershift assay was performed with adding 2 µg of specific anti-c-Jun antibody prior to adding labeled AP-1 oligonucleotides. NS; non-specific binding.

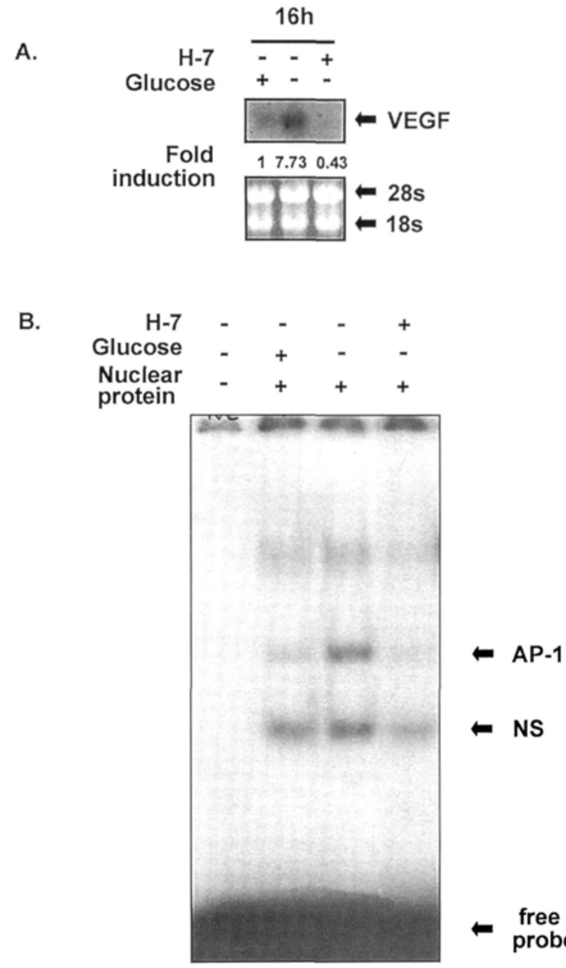

Figure 2. PKC was involved in elevated VEGF expression and AP-1 DNA-binding activity under hypoglycemia. A, HepG2 cells were treated with 60 µg/ml of H-7 and incubated in normal or glucose-free medium for 16 h. Total RNA (30 µg) from cells was electrophoresed and Northern blotting was performed. B, after cells were treated with H-7 and incubated for 1 h, nuclear extracts were isolated and EMSA was performed.

hypoglycemia-induced PKC activity enhances mRNA expression of c-*jun* (23). Based on these results, we reasoned that hypoglycemia-elevated PKC activity can influence VEGF expression and AP-1 DNA-binding activity. HepG2 cells were treated with 60 µg/ml of H-7, a specific PKC inhibitor for 16 h for Northern blotting or 1 h for EMSA under hypoglycemic condition. As shown in Fig. 2, the enhanced VEGF expression and AP-1 DNA-binding activity caused by glucose-deprivation were completely abrogated by the treatment of H-7. These results indicate that PKC plays a crucial role on hypoglycemia-induced VEGF expression as well as AP-1 activity.

Increased intracellular Ca^{2+} levels under hypoglycemic condition are involved in hypoglycemia-enhanced VEGF expression and AP-1 DNA-binding activity. Although it has been reported that hypoglycemia increases intracellular Ca^{2+} in neuronal cells (24) as well as in insulinoma (25), there is no report on Ca^{2+} increase by hypoglycemia in tumor cells. Moreover, intracellualr Ca^{2+} concentration influences VEGF expression (16) as well as AP-1 DNA-binding activity (26), but little is known about Ca^{2+} involvement in hypoglycemia-induced VEGF and AP-1 activity. Therefore, we measured intracellular Ca^{2+} level of HepG2 cells under the glucose-free condition. We resuspended Fura-2/AM-loaded HepG2 cells in calcium-free KRB with or without 6 mM glucose and surprisingly we found out that intracellular Ca^{2+} of HepG2 cells in glucose-free KRB showed a continuous increase of Ca^{2+} concentration for 40 min about six-fold that of cells in glucose-containing KRB (Fig. 3A). Furthermore, to elucidate a role of Ca^{2+} on the VEGF expression and AP-1 DNA-binding activity under hypoglycemia, we treated HepG2 cells exposed to hypoglycemia with 100 nM of BAPTA/AM, an intracellular Ca^{2+} chelator for 16 h for Northern blot analysis or 1 h for EMSA. As shown in Fig. 3B and C, BAPTA/AM markedly reduced hypoglycemia-increased VEGF expression and AP-1 activity, suggesting the existence of correlation between intracellular Ca^{2+} and hypoglycemia in HepG2 cells.

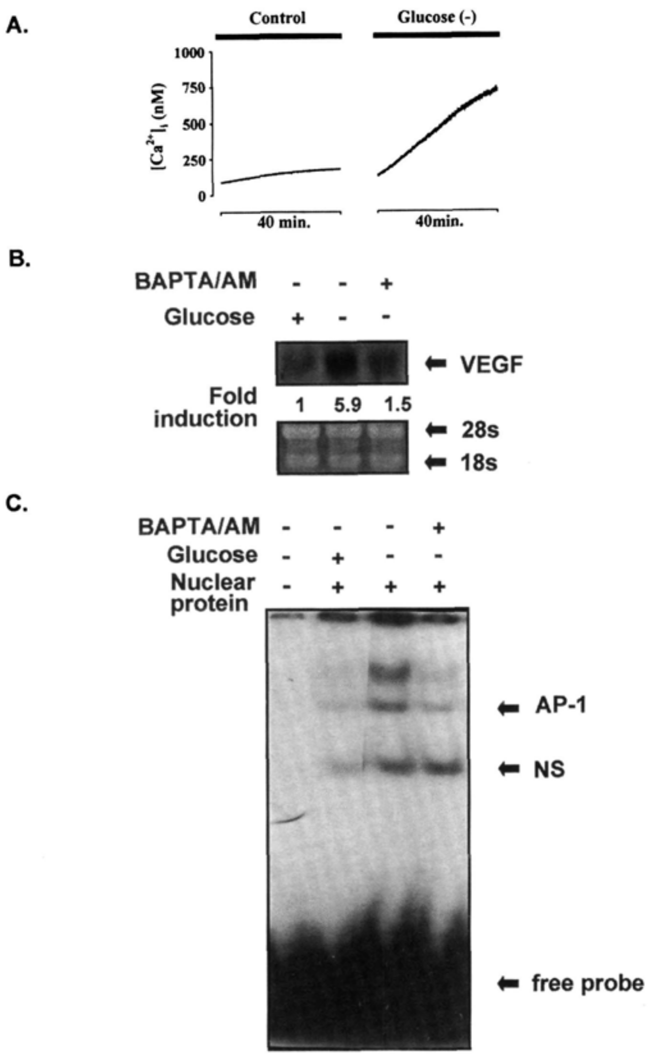

Figure 3. Intracellular Ca^{2+} in elevated VEGF expression and AP-1 DNA-binding activity under hypoglycemia. A, cells were resuspended in Ca^{2+}-free KRB with or without 6 mM glucose. Data represented intracellular Ca^{2+} changes for 40 min. B, cells were pretreated with 0.1 μM BAPTA/AM for 4 h. After incubating cells under hypoglycemic condition for 16 h, Northern blotting was performed. C, 0.1 μM BAPTA/AM was pre-treated in HepG2 cells for 4 h in normal condition and additionally incubated for 1 h in glucose-free medium, and EMSA was performed.

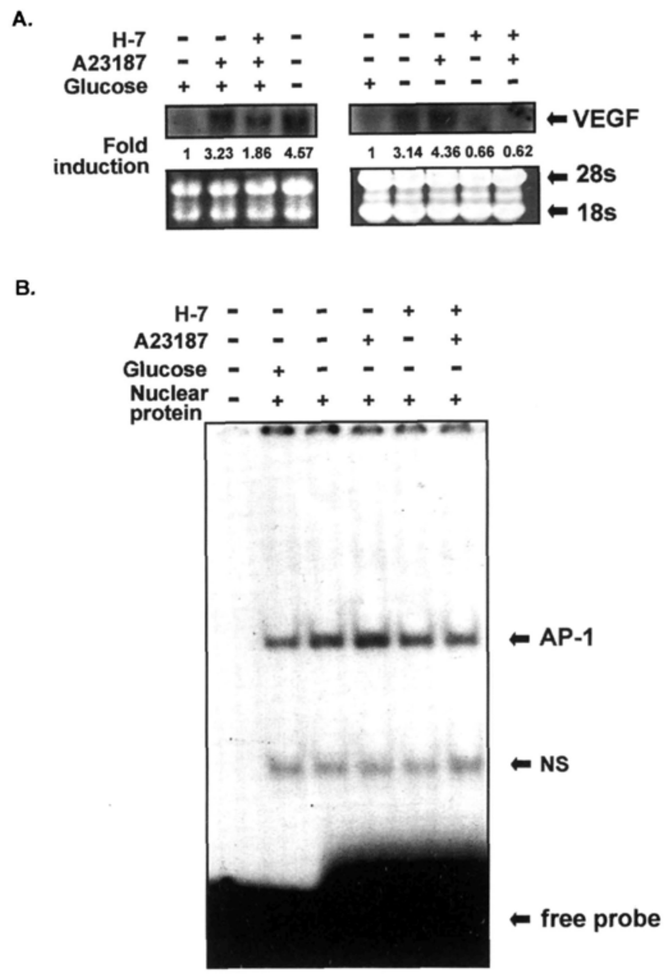

Figure 4. Roles of Ca^{2+} and PKC on hypoglycemia-induced VEGF expression and AP-1 DNA-binding activity. A, HepG2 cells were pre-treated with 50 nM of A23187 for 4 h, and then incubated in the absence or in the presence of 60 μg/ml H-7 for 8 h under hypoglycemic condition. Total RNA from cells was electrophoresed and Northern blotting was performed. B, after cells were pretreated with 50 nM A23187 for 4 h, 60 μg/ml H-7 was treated for 1 h under hypoglycemia and EMSA was performed.

PKC is involved in the Ca^{2+}-mediated VEGF expression and AP-1 DNA-binding activity under hypoglycemia. Although Ca^{2+} is known to modulate PKC activity (20) and PKC activation was induced by hypoglycemia (23), a role of Ca^{2+} associated with PKC in hypoglycemic induction of VEGF and AP-1 DNA-binding activity was not reported. Therefore, to delineate whether increased VEGF expression and AP-1 DNA-binding activity by hypoglycemia could be regulated by Ca^{2+}-dependent PKC activity, we co-treated HepG2 cells with Ca^{2+} ionophore A23187 and/or H-7 (Fig. 4). HepG2 cells were incubated with 50 nM of A23187 for 8 h in combination with or without 60 μg/ml of H-7 under normal or hypoglycemic condition. As expected, induced VEGF expression due to increased Ca^{2+} by A23187 under normal condition was significantly reduced by H-7 (Fig. 4A, lane3). Accordingly, hypoglycemic induction of VEGF was more increased by A23187 (lane 7) whereas co-treatment with H-7 completely abolished the VEGF induction (lane 9). In addition, more increased AP-1 activity in combination of A23187 with glucose deprivation was significantly blocked by H-7 (Fig. 4B, lanes 4, 6). Therefore, both Ca^{2+} and PKC are apparently involved together in hypoglycemic induction of VEGF as well as AP-1 activity.

Discussion

Several researchers have shown that the malignancy of solid tumors is dependent on neovascularization near the tumor, which is necessary to provide adequate nutrients and oxygen (27), but the study of angiogenesis has been mainly progressed in terms of hypoxia although nutrient depletion may play an important role on angiogenesis. In addition, the mechanism that mediates an increase of VEGF expression and AP-1 activity is not fully understood, especially to hypoglycemia.

In this study, to elucidate controlling mechanism of hypoglycemia-induced VEGF expression, we determined a time-dependent induction of VEGF mRNA in HepG2 cells and enhancement of AP-1 DNA-binding activity in HepG2 cells cultured under hypoglycemic condition (Fig. 1). Our observations suggest that the increased AP-1 activity by hypoglycemia may transactivate the VEGF gene.

PKC activity is increased by hypoglycemia in MCF-7/ADR cells (23) and PKC mediates stress-induced VEGF expression (15). We showed that induction of VEGF under hypoglycemia was reduced by H-7 (Fig. 2A), and concomitantly diminishing AP-1 activity (Fig. 2B). These results strongly suggest the involvement of PKC in the regulation of DNA-binding activity of AP-1 under hypoglycemia.

Study for hypoglycemia until now has focused on neuronal cells because brain uses glucose only as its energy source (28). When neuronal cells are exposed to hypoglycemia, intracellular Ca^{2+} is increased (7). Ca^{2+} levels of mammalian cells are between 100-500 nM in resting state (23). As shown in Fig. 3A, intracellular Ca^{2+} levels of HepG2 cells in the resting stage were approximately 100-200 nM, and slightly increased within 40 min but did not run over 500 nM. Under hypoglycemia, however, Ca^{2+} levels increased markedly and continuously, and did not arrive at the plateau within 40 min, indicating that intracellular Ca^{2+} might be deeply associated with hypoglycemia in tumor cells. A Na^+/Ca^{2+} exchanger blocker (21), benzamil diminished hypoglycemia-increased Ca^{2+} concentration (data not shown), suggesting that a kind of Ca^{2+} source, under hypoglycemia at least, may be a Na^+/Ca^{2+} exchanger. In addition, we treated HepG2 cells with BAPTA/AM to investigate the role of intracellular Ca^{2+} on the induction of VEGF expression and AP-1 activity under glucose deprivation (Fig. 3B and C). Chelation of intracellular Ca^{2+} suppressed hypoglycemia-induced VEGF expression and hypoglycemia-stimulated AP-1 DNA-binding activity, which is consistent with our hypothesis that intracellular Ca^{2+} regulates both the hypoglycemic induction of VEGF expression and the AP-1 activity. We also showed that PKC might be involved with an increase of intracellular Ca^{2+} levels in hypoglycemia-enhanced VEGF expression. As shown in Fig. 4, VEGF expression and AP-1 activity increased by A23187 in hypoglycemia and were completely abolished by H-7. Although we cannot rule out other functions of H-7 as a general kinase inhibitor, the concentration of H-7 applied in this study was relatively specific for PKC (23).

Our results provided evidence that the increase of intracellular Ca^{2+} level induces the activation of PKC, which induce the activation of AP-1 leading to the increase of VEGF in glucose-deprived environment. These findings have similarities to VEGF regulation by hypoxia (7,22), suggesting that as tumors grow, they may be exposed to the depletion of oxygen and nutrients simultaneously. Growing tumor cells may share a large body of the signal transduction pathway for these stresses. Hypoglycemic microenvironmental stress in tumor cells may regulate angiogenesis by controlling the release of angiogenic factors such as VEGF through changing intracellular Ca^{2+} levels and activity of transcription factor during tumor pathogenesis.

Acknowledgments

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